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MASTER THESIS

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Abstract

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Chapter 1

Introduction

Multilevel models are a standard statistical tool for analysing clustered data, which frequently occur in the social sciences (e.g., patients nested within therapists, students within classes, or employees within firms). In practice, such data sets are often characterized by relatively small numbers of groups and by the presence of missing values.

Methodological research has proposed a range of approaches to address each of these challenges separately. Small-sample corrections for multilevel models have been developed to improve statistical inference when the number of clusters is limited (Elff et al., 2021; D. M. McNeish & Stapleton, 2016). At the same time, modern missing-data methods have been extended to multilevel data structures (Enders, 2025).

However, relatively little research has investigated how these two challenges interact. In particular, it remains relatively unclear how small-sample inference procedures perform when model parameters are estimated in the presence of missing data and handled using modern missing-data methods.

This thesis investigates this interaction. Specifically, it examines whether applying small-sample-corrected degrees of freedom after multiple imputation improves the inferential quality of estimates in random intercept models, and whether fully Bayesian estimation provides a viable alternative approach.

The thesis first introduces the theoretical foundations of multilevel models and reviews current methodological approaches for handling small samples and missing data. Subsequently, a simulation study is conducted to investigate the performance of these methods under different data conditions.

Chapter 2

Theory

2.1 Theoretical foundation of multilevel models

Example

Imagine you are interested in the musical self-concept of young adults and whether it is related to how many hours they practice their instrument each week.
To investigate this question, you collect data from several youth ensembles. You recruit 15 ensembles, each consisting of approximately 10 players. Every player completes a questionnaire assessing their musical self-concept as well as their weekly practice time.
At first glance, this seems like a simple dataset of 150 individuals. But is it really that simple?
Players from the same ensemble rehearse together, share teachers, and experience similar social and musical environments. Their responses may therefore be more similar within ensembles than between ensembles.

Populations like this are common in the social sciences. Units of interest (e.g., students, employees, musicians) are grouped into higher-level units such as classes, firms, or ensembles. The resulting data is called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7).
From a sampling perspective, a so-called *multistage sampling process* (sample groups first, then individuals inside these groups[1]) is not independent: After selecting a group, you will try to sample all of the members of this group, making it more likely to select one person in this group over others in non-selected groups (Snijders & Bosker, 2012, p. 7). More importantly, individuals inside the same group typically share contextual influences (e.g., the same ensemble teacher, the same class room) and thus will give more similar responses in surveys and behavior. As a consequence, the basic assumption of independence in traditional linear modeling is violated, because error terms will be correlated (Raudenbush & Bryk, 2010, p. xx).
One well-suited and popular possibility to account for this fact is the use of *multilevel models* (also called *hierarchical linear models* or *mixed models*).

[1] In this thesis, the term 'groups' refers to the higher-level units of the data and level-1 units are referred to as 'persons'. However, it is important to note that these models can also be used for longitudinal research, where measurement points are on level 1 and individuals are the level-2 units. Furthermore, more than two levels are also possible, but are not discussed here.

2.1.1 The concept of multilevel regression models

The aim of regression analysis, whether single- or multilevel, is to explain variance in an outcome variable by including associated variables in the model as predictors (Snijders & Bosker, 2012, p. 80). In single-level regression, the residual e_i captures the *unexplained variance* in Y, that is differences in the outcome from the prediction that the model made (Leonhart, 2013, p. 319). The assumption of the classical linear model is that after inclusion of all *systematic* influences on Y, the residuals are *random*. Specifically, they are assumed to have expectation zero ($E(e_i) = 0$), constant variance across observations (homoscedasticity; $Var(e_i) = \sigma^2$), and to be independent across observations ($Cov(e_i, e_{i'}) = 0$ for $i \neq i'$) (Fahrmeir et al., 2021, p. 87).

Multilevel models (MLM) extend regular linear regression by adding level-2 *random effects* to explain the variability between groups that cannot be accounted for by predictors (Snijders & Bosker, 2012, p. 80). This partition of variance can be expressed in the multilevel regression model without predictors, the *empty model*:

$$Y_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

Here, the outcome value of person i in group j is modeled as the sum of the overall mean γ_{00}, a group level random effect u_{0j} and a random effect on the individual level, e_{ij} (Snijders & Bosker, 2012, p. 49). The random intercept u_{0j} captures the dependency of individuals within one group statistically, as all individuals in group j have the same u_{0j} value and their outcome values are therefore correlated. Conditional on this group component, however, the remaining level-1 residuals e_{ij} are assumed to be independent, addressing the issue of dependency that would arise if a single-level regression model were applied to nested data.

This and the models explained below are illustrated in Figure 2.1.

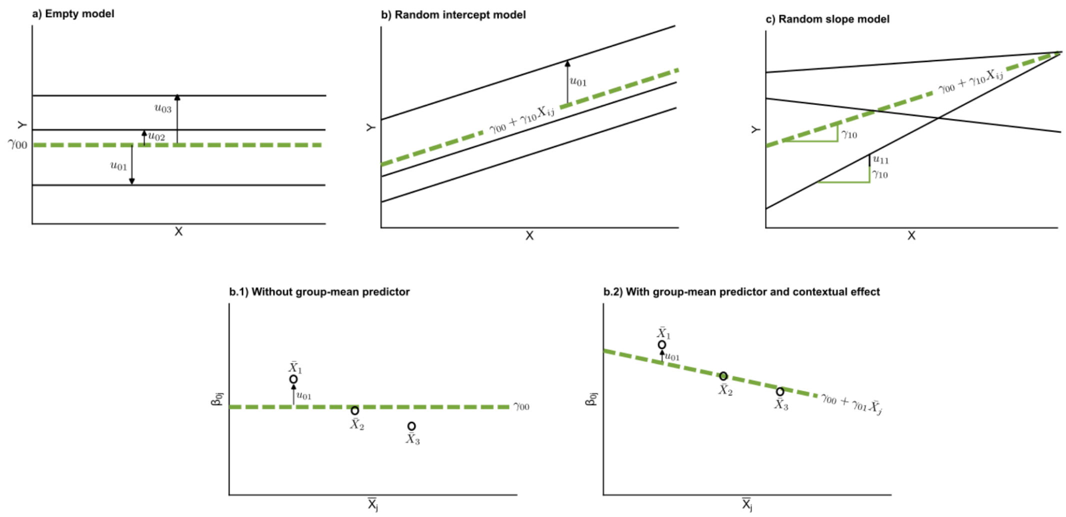


FIGURE 2.1: Illustration of common multilevel model specifications.

Note. Top row: empty model, random intercept model, and random slope model. Bottom row: group-level relationships between group means of the predictor $\bar{X}_{.j}$ and the group-specific intercepts β_{0j}.

2.1.2 Within- and between- group regressions and inclusion of predictors

Predictors can then be included to explain variability within groups (level 1) and between groups (level 2).
At level 1, the outcome of a person i in group j is modeled as a function of individual-level explanatory variables X_{ij} to obtain the within-group regression model. For simplicity, the model is illustrated with a single level-1 predictor:

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + e_{ij} \tag{2.1}$$

At level 2, the variation between groups in the group-dependent coefficients β_{0j} and β_{1j} is modeled. Each group may have its own intercept and slope. These coefficients can be separated into a fixed part (the average coefficient across groups) and a random deviation from this average (Snijders & Bosker, 2012, p. 75). Without level-2 predictors:

$$\beta_{0j} = \underbrace{\gamma_{00}}_{\text{Fixed part}} + \underbrace{u_{0j}}_{\text{Random part}}$$

$$\beta_{1j} = \underbrace{\gamma_{10}}_{\text{Fixed part}} + \underbrace{u_{1j}}_{\text{Random part}}$$

On this level, in order to explain between-group variability in the intercepts and slopes, level-2 predictors Z_j may be included (Snijders & Bosker, 2012, p. 80). Including such predictors may reduce the unexplained variance of the random effects u_{0j} and u_{1j}. With a single level-2 predictor, the model becomes:

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j} \tag{2.2}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j} \tag{2.3}$$

Substituting the level-2 equations Equation 2.2 and Equation 2.3 into the level-1 model Equation 2.1 yields the combined multilevel model:

$$Y_{ij} = \underbrace{\gamma_{00} + \gamma_{10}X_{ij} + \gamma_{01}Z_j + \gamma_{11}X_{ij}Z_j}_{\text{Fixed part}} + \underbrace{u_{0j} + u_{1j}X_{ij} + e_{ij}}_{\text{Random part}}$$

Y_{ij}: Outcome for individual i in group j
X_{ij}: Level-1 (within-group) predictor
Z_j: Level-2 (between-group) predictor

γ_{00}: Fixed intercept (overall mean when $X_{ij} = 0$, $Z_j = 0$)
γ_{10}: Average within-group effect of X_{ij} on Y_{ij}
γ_{01}: Effect of Z_j on the intercept (between-group effect)
γ_{11}: Cross-level interaction; change in the effect of X_{ij} depending on Z_j

u_{0j}: Random intercept (group-specific deviation from γ_{00})
u_{1j}: Random slope (group-specific deviation from γ_{10})
e_{ij}: Level-1 residual

If no level-2 predictors are included ($Z_j = 0$), the model reduces to the simpler random-coefficients model without contextual predictors.

2.1.3 Distributional assumptions of the multilevel regression model

Similar distributional assumptions are made for the level-2 random effects as for the residuals in single-level regression: they have mean zero, constant variances across groups, and are independent across clusters and independent of the level-1 residuals. However, random effects within the same cluster may be correlated and are often expected to do so (Snijders & Bosker, 2012, p. 76). Additionally, residuals and random effects are commonly assumed to follow a normal distribution for convenience, as it facilitates the likelihood-based estimation of the unknown parameters (Fahrmeir et al., 2021, p. 388). In summary:

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{T} \right), \qquad \mathbf{T} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix}$$

$$e_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$$\mathrm{Cov}(u_{0j}, e_{ij}) = 0, \quad \mathrm{Cov}(u_{1j}, e_{ij}) = 0$$

2.1.4 The random intercept model

In many empirical applications of multilevel models the slope β_{1j} is assumed to be constant across groups ($u_{1j} = 0$), leading to the *random intercept model* (Snijders & Bosker, 2012, p. 45). This model allows for baseline differences in the outcome between groups, whereas the strength of the relationship between predictors and the outcome is assumed to be constant:

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + \gamma_{01} Z_j + u_{0j} + e_{ij}$$

The random intercept model also provides a convenient framework for examining the use of *group means* of a given level-1 predictor as additional level-2 predictors. Including the group mean as a contextual variable allows the within-group regression coefficient to vary from the between-group regression coefficient (Snijders & Bosker, 2012, p. 57). This makes the model more flexible and often more realistic, as so-called *contextual effects* are frequently observed in empirical settings (Snijders & Bosker, 2012, p. 27). In such cases, the effect of an individual-level predictor differs from the effect of the average value of this predictor. The following example illustrates how processes operating within ensembles may differ from those operating between ensembles.

Example

Students often compare their own practice behavior with that of other ensemble members.
A student who practices more than their peers may feel more competent and report a higher musical self-concept.
At the same time, students in ensembles where everyone practices a lot may feel less competent because upward social comparisons become more likely.
This leads to two possible hypotheses:

- Students who practice more than others in their ensemble tend to report a higher musical self-concept.

- Students in ensembles with a higher average practice time may report a lower musical self-concept.

To separate within- and between-group effects, the level-1 predictor is then often *group-mean-centered*: Instead of using X_{ij} directly, the centered predictor $(X_{ij} - \bar{X}_j)$ is entered into the model. This decomposes within- and between- group effects and facilitates their interpretation (Snijders & Bosker, 2012, p. 58). A random intercept model including such a decomposition can be written like this:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + u_{0j} + e_{ij} \tag{2.4}$$

In this random intercept model with a contextual effect, the musical self-concept of student i in ensemble j is predicted by the sum of several components:

- γ_{00}: the average musical self-concept across all ensembles
- $\gamma_{10}(X_{ij} - \bar{X}_{.j})$: the effect of the student's practice time relative to the average practice time in their ensemble
- $\gamma_{01}\bar{X}_{.j}$: the effect of the ensemble's average practice time
- u_{0j}: a random deviation of ensemble j's average musical self-concept from the overall mean (unexplained differences between ensembles)
- e_{ij}: a random deviation of the student's musical self-concept from the predicted value within their ensemble

The regression coefficients represent two conceptually different effects:

- γ_{10}: the *within-group effect*, indicating how a student's own practice time relates to their musical self-concept while controlling for the ensemble context
- γ_{01}: the *contextual effect*, indicating how the ensemble's average practice time relates to students' musical self-concept

For the remainder of this thesis, the focus lies on this particular random intercept model.

2.1.5 The intraclass correlation coefficient (ICC)

In the random intercept model, the variance components are given by

$$u_{0j} \sim \mathcal{N}(0, \tau_0^2), \qquad e_{ij} \sim \mathcal{N}(0, \sigma^2), \qquad \mathrm{Cov}(e_{ij}, u_{0j}) = 0.$$

From these variance components, the proportion of variability that is attributable to between-group differences can be calculated. This quantity is referred to as the *intraclass correlation coefficient* (ICC). In the empty model, the ICC is equal to the correlation between the outcomes of two randomly chosen individuals from the same group (Snijders & Bosker, 2012, p. 18). In models including predictors, the ICC is interpreted conditionally on the predictors, that is, as the correlation between two individuals from the same group who share identical predictor values (also referred to as the *residual ICC*) (Snijders & Bosker, 2012, p. 52).

Formally, the ICC is defined as

$$ICC = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}$$

The ICC is an important diagnostic metric because it indicates whether the use of a multilevel model is advantageous. If ICC = 0, no between-group variance is present and a conventional single-level regression model would typically be sufficient. Conversely, larger ICC values indicate increasing similarity within groups and thus statistical dependency of observations. Ignoring this dependency and applying a classical linear model leads to biased standard errors, confidence intervals, and hypothesis tests (Fahrmeir et al., 2021, p. 370).

2.2 Estimation of multilevel models

In the preceding chapter, the theoretical foundations and parameters of the multilevel model were introduced. In practice, however, these parameters are unknown and only exist in the theoretical realm as 'true' values. The fixed effects γ and variance components σ^2 and τ^2 must therefore be estimated from observed data. Based on these estimates, inferences about the underlying population parameters are drawn. This process is illustrated for the random intercept model in Figure 2.2. Provided that the model specification corresponds to the true data-generating process well enough, the reliable performance of this 'inferential machinery' depends on three main elements (Elff et al., 2021):

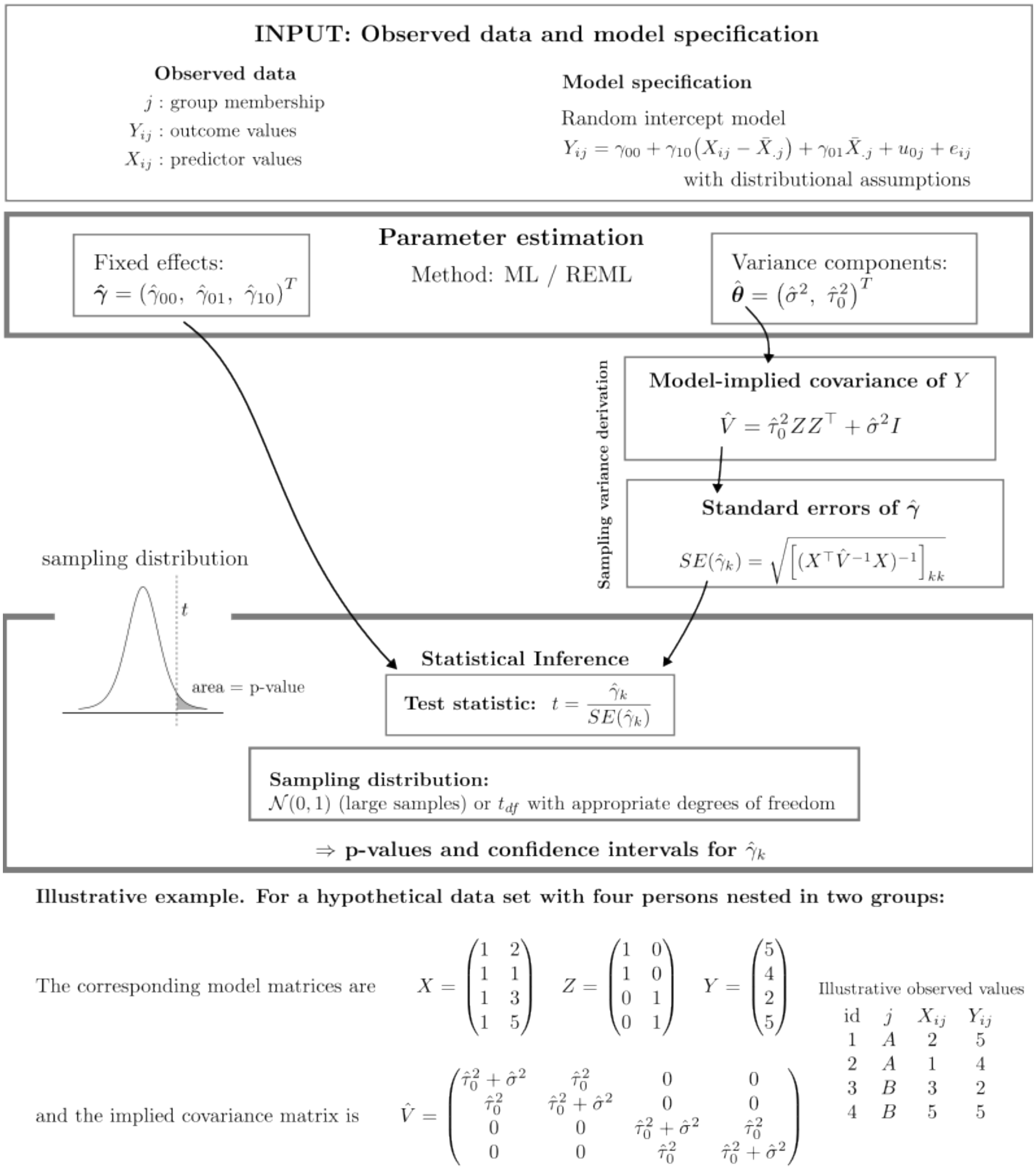
1. Estimation of fixed-effect coefficients (regression parameters)
2. Estimation of variance–covariance components of the random effects (u) and residual errors (e)
3. Selection of an appropriate approximation of the sampling distribution of the test statistic.

Aspects of the observed data may pose challenges to this estimation process. Moreover, the inferential machinery itself (i.e., its formulas and underlying assumptions) may be inapt for the empirical data reality. Two common challenging aspects, namely small sample sizes and missing data, will be discussed in the following chapters.

2.3 Small sample sizes in multilevel models

Frequentist estimation procedures for linear multilevel models typically rely on large-sample (asymptotic) theory. In multilevel models, the number of groups is particularly important. Simulation studies suggest that approximately 25–30 clusters are often required for reliable estimation, depending on the model specification and the parameters of interest (D. McNeish, 2017; D. M. McNeish & Stapleton, 2016). In practice, however, such numbers are often difficult to achieve because recruiting a large number of groups can be challenging and costly.

When the number of groups is small, several components of the inferential procedure may perform poorly, particularly the estimation of variance components, the calculation of standard errors, and the approximation of sampling distributions. The main issues associated with small-sample multilevel models and available corrections and alternatives are presented below.



Simplified frequentist inferential process in a random intercept model, based on Snijders & Bosker (2012) and Raudenbush & Bryk (2010).

Notes on notation

- X: design matrix for the fixed effects. Each row corresponds to an observation (e.g., a person) and each column to a predictor included in the fixed part of the model.
- Z: design matrix for the random effects. In the random intercept model it encodes group membership (1 = observation belongs to the group, 0 = otherwise), so observations from the same group share the same random intercept.
- Y: outcome vector containing the observed values of the dependent variable.
- $\hat{V}$: estimated variance–covariance matrix of the outcome vector Y. In the random intercept model it is determined by the estimated variance components of the random intercept ($\hat{\tau}_0^2$) and the residual variance ($\hat{\sigma}^2$), reflecting that observations within the same group are correlated.
- $\hat{\gamma}_k$: the k-th element of the estimated fixed-effect vector $\hat{\gamma}$.

FIGURE 2.2

2.3.0.1 Estimation of regression parameters

The regression parameters are estimates of the effects of interest. For example, how much does one additional hour of weekly practice influence a student's musical self-concept? How does the musical self-concept differ between ensembles that practice on average three hours per week and those that practice thirteen hours per week?
Kackar & Harville (1981) showed that both ML and REML estimators of variance components are even and translation-invariant, implying that fixed effect point estimates are unbiased even in small samples.

2.3.0.2 Estimation of variance components

The estimation of the variance components is more delicate. In ML estimation, fixed effects and variance components are estimated simultaneously, typically via iterative procedures like the Expectation Maximization (EM) mechanism (Daniel McNeish, 2017). In this process fixed effects are effectively being treated as known when estimating variance components. Consequently, the sampling variability of the fixed-effect estimates is ignored and the loss of degrees of freedom due to their estimation is not accounted for. While these issues are not troublesome in large samples, they lead to systematic underestimation of variance components in small samples (Daniel McNeish, 2017).
REML addresses this issue by separating the estimation conceptually. The fixed effects are eliminated from the likelihood, and variance components are estimated solely based on residual information. Fixed effects are then estimated via generalized least squares (GLS), using the previously estimated variance components (Raudenbush & Bryk, 2010, p. 410). This leads to less downward-biased variance component estimates, while ML and REML become asymptotically equivalent as sample size increases.

Nevertheless, the the estimated variance of the fixed-effect estimators may still be too small in small samples for two reasons. First, the variance formula treats the estimated variance components $\hat{\theta}$ as known and therefore ignores their sampling variability (Kackar & Harville, 1984). Second, the REML variance expression itself relies on asymptotic approximations (Daniel McNeish, 2017). Both issues can lead to underestimated standard errors. Several small-sample corrections have been proposed to address this problem, among which the Kenward–Roger procedure is one of the most widely used (Daniel McNeish, 2017).

2.3.0.3 Approximation of the sampling distribution

The standardized test statistic used for inference (see Figure 2.2) can be regarded as 'small-sample-adjusted' at this point. The final step concerns the choice of a sampling distribution for p-values and confidence intervals. In small samples, a t-distribution is typically recommended instead of a standard normal distribution (Snijders & Bosker, 2012, p. 95).

The shape of the t-distribution is determined by the number of degrees of freedom. With small degrees of freedom, even small changes noticeably affect the shape of the sampling distribution and therefore the resulting p-values and confidence intervals. The concept of degrees of freedom is difficult to define concisely, but it can be described as the number of independent pieces of information in the data that remain after accounting for restrictions such as estimated parameters (Pandey & Bright, 2008).

In multilevel models, the degrees of freedom generally cannot be calculated exactly and must therefore be approximated because of the nested data structure (Snijders &

Bosker, 2012, p. 95). The Kenward–Roger correction mentioned earlier also includes a complex small-sample-adjusted approximation of the degrees of freedom, building on the Satterthwaite method (Daniel McNeish, 2017).

A more straightforward approximation is provided in textbooks by Raudenbush & Bryk (Raudenbush & Bryk, 2010, p. 58) and Snijders & Bosker (Snijders & Bosker, 2012, p. 95), and has also been proposed in methodological articles (Elff et al., 2021). This approach distinguishes between level-1 and level-2 predictors and reflects the concept of degrees of freedom as information minus estimated parameters:

$$df_{Lv1} = N_1 - r - 1, \qquad df_{Lv2} = N_2 - q - 1$$

N_1 : Level-1 sample size (total number of individuals)
r : Total number of predictors
N_2 : Level-2 sample size (number of groups)
q : Number of predictors at level two

2.3.1 Fully Bayesian analysis

As alternative to the frequentist framework predominantly presented here, a Bayesian approach is often recommended for small sample sizes (Daniel McNeish, 2016a). In this framework, parameters are viewed as random variables with probability distributions. Prior knowledge about the parameters of interest in the form of prior distributions is combined with the information in the observed data in the form of a likelihood function. Bayes' theorem yields the posterior distribution, representing the updated knowledge about the parameters (Van De Schoot et al., 2014). More general details on Bayesian analyses will not be discussed in this thesis.

Two aspects of Bayesian analyses are particularly helpful in small samples.
First, Bayesian inference does not rely on large-sample asymptotic approximations. Instead, the posterior distribution, the end result of a Bayesian analysis, is typically approximated using Markov Chain Monte Carlo (MCMC) sampling. Consequently, Bayesian inference does not require small-sample corrections for standard errors or degrees of freedom.
Second, Bayesian methods allow the incorporation of prior information about parameters. Informative or weakly informative priors may improve estimation accuracy by adding information when the likelihood is relatively flat in small samples (Gelman et al., 2025, p. 355). At the same time, prior specification becomes particularly influential under such conditions. Vague or software default priors may therefore have unintended effects and can perform poorly in small samples (Daniel McNeish, 2016a, 2016b; Gelman, 2006). Nevertheless, Bayesian estimation has in some settings been found to outperform maximum likelihood estimation in terms of RMSE, even when noninformative priors are used (Zitzmann et al., 2015). In addition, prior distributions may prevent inadmissible parameter estimates that could otherwise lead to convergence problems, such as negative variance estimates (Van De Schoot et al., 2014). Thus, the inclusion of prior distributions may be regarded either as a methodological advantage or as "a price to be paid" (Raudenbush & Bryk, 2010, p. 411) for Bayesian inference.

2.4 Missing data in multilevel models

> Example
>
> In the study about musical self-concept, you find after data collection that some students in several ensembles did not report how many hours they practice per week. They completed the questionnaire on musical self-concept but left the practice question blank.
> The reason for this is unknown: they might have forgotten to answer the question, skipped it intentionally, or misunderstood it. Regardless of the reason, the data set now contains missing values. How should these missing observations be handled in the analysis?

Missing data (i.e., incomplete data sets) are a widespread issue in empirical research and of course also applies to multilevel data, e.g., because individuals drop out of studies or forget to fill out certain items of a questionnaire. Three major concerns commonly arise from data that contain missing values (Lüdtke et al., 2007):

1. The effective sample size is reduced, leading to lower efficiency in estimation (this is especially troublesome if the would-be sample size is already small)
2. Typical statistical methods are difficult to conduct, because they were designed for complete data.
3. Parameter estimation may be biased, if the missing and the observed data differ from each other systematically

Therefore, it is crucial to treat missing data appropriately. In the last three decades, this issue has received attention in psychological research, and modern methods were developed (Enders, 2022; Schafer & Graham, 2002), more recently also for the more complex case of multilevel data (Enders, 2025; Lüdtke et al., 2017). This chapter discusses these methods, after giving a short introduction into different missing data mechanisms and patterns.

2.4.1 Missing data mechanisms

Why data are missing can have multiple reasons, and different mechanisms may lead to different consequences in statistical inference. Rubin and Little (Little & Rubin, 1987; Rubin, 1976) introduced a typology and theoretical framework of missing data mechanisms that describes the relationship between the observed and the missing data. The missing data mechanism is generally not empirically testable from the observed data alone and assumptions about the mechanism must therefore be theoretically justified (Enders, 2022, p. 14ff.).

Missing completely at random (MCAR). Considering the hypothetical complete data matrix Y[2], Y_{com} is partitioned into observed, Y_{obs}, and missing parts, Y_{mis}, and R denotes the missingness[3]. Missing values are considered to be MCAR if the distribution of missingness does not depend on the data (Schafer & Graham, 2002):

$$P(R \mid Y_{com}) = P(R)$$

[2] Here, Y represents the complete data set, including both outcome variables and predictors.

[3] R is treated as a set of random variables that indicate for each value if it is missing or not (Schafer & Graham, 2002).

In other words, no variable (observed or unobserved) can predict missingness and every individual in every group has the same chance of missing values (Enders, 2022, p. 6).

Missing at random (MAR). The MAR mechanism describes the case when the probability of missingness does depend on observed data, but not on missing values (Enders, 2022, p. 8):

$$P(R \mid Y_{com}) = P(R \mid Y_{obs})$$

Missingness is therefore not random, as the name would suggest, but *conditionally random*: After controlling for the observed data, the missingness is purely random (Enders, 2022, p. 8).

For example, students who practice less might be more likely to skip the question about weekly practice duration. If this behavior can be predicted from other observed variables (musical self-concept or ensemble membership), the MAR assumption could be plausible.

Missing not at random (MNAR). If missingness depends on unobserved values, this is called a missing not at random process[4]:

$$P(R \mid Y_{com}) = P(R \mid Y_{obs}, Y_{mis})$$

Two individuals in the same group with identical scores in all observed variables therefore no longer have the same chance of missing values, because this chance depends on the unavailable, missing information itself (Enders, 2022, p. 11).

While these mechanisms also apply to multilevel data, the situation becomes more complex because missing values can occur in the outcome, in level-1 predictors, in level-2 predictors, or even in the grouping variable (or combinations of these; Buuren (2021)).

2.4.2 Methods for handling missing data

Many estimation procedures require complete data. Therefore, missing data must be handled explicitly before model estimation. If researchers do not actively specify a method, statistical software often defaults to using only complete observations (Lüdtke et al., 2007). This simple ad hoc procedure is called *listwise deletion* (LD) or *complete case analysis* and is a widely used approach to handling missing data. Cases with at least one missing value are discarded, and the model is estimated using only complete observations (Graham, 2009). In multilevel settings, missingness on the group level implies that all individuals in this group are excluded as well.
Listwise deletion has two major disadvantages (Enders, 2022, p. 24):

- Because partial data is unused, statistical power is reduced.

- If the missing data mechanism is not MCAR, parameter estimates may be highly biased.

LD can still be adequate under some conditions (Enders, 2022, p. 25; Graham, 2009), but in already small samples, additional data deletion should be avoided as power is already limited (D. McNeish, 2017).

[4]The special case $P(R \mid Y_{com}) = P(R \mid Y_{mis})$ is referred to as **focused MNAR** (Enders, 2022, p. 11).

More modern and better-performing methods are available and will be introduced in the following sections. These approaches generally rely on the assumption that data are missing at random (MAR). Methods for dealing with MNAR processes are not discussed in this thesis.

Modern approaches to missing data can be broadly specified into two classes:

1. *Model-based approaches* to missing data specify a statistical model for the complete data and base inference on the likelihood or posterior distribution implied by that model. Missing values are handled directly within the estimation procedure (Little & Rubin, 2020, p. 25).
2. *Imputation-based approaches* are two-fold: First, substitutions for the missing values are estimated, leading to complete data sets, which are then analyzed using standard methods.

2.4.2.1 Multiple Imputation (MI)

Multiple imputation is an imputation-based method, as the name suggests. The basic idea is to replace missing values with plausible draws (‘imputations’) from their predictive distribution. The substantive analysis model is then fitted to the imputed data. In order to include the uncertainty due to missing values, in MI several imputations for each missing observation are generated. By incorporating the variance between these imputations, the additional uncertainty can be captured (Enders, 2022, p. 283). The analysis of incomplete data is separated into three steps: imputation, analysis, and pooling.

2.4.2.1.1 Creating imputations

Imputations can be understood as “predicted values plus noise” (Enders, 2022, p. 266). Multiple imputation is closely related to Bayesian data augmentation techniques, in which missing values are treated as additional unknown quantities (Tanner & Wong, n.d.). From the joint distribution of regression parameters θ and the missing values $P(\theta, Y_{mis} \mid Y_{obs})$, parameters and imputations are simulated iteratively by an MCMC algorithm (Enders, 2022, p. 190). While during this process many thousand posterior draws are generated, only a small number (usually between $m = 20$ and $m = 100$ is recommended, Graham et al. (2007)) of imputations is saved, leading to m complete data sets. The posterior draws of the parameters are discarded (Gelman et al., 2025, p. 451).
The distribution of missing values (i.e., the imputation model) is either conceptualized jointly for all variables (*joint imputation*) or individually (*fully conditional specification*). Both approaches are also suitable for multilevel data structures (Enders et al., 2016). In joint modeling, a single multivariate distribution for all variables with missing values is specified, often the multivariate normal distribution in case of continuous variables (Enders, 2022, p. 263). *Fully Conditional Specification* approximates the joint distribution by cycling through a sequence of univariate models for each variable with missings, conditional on the others. This approach is suitable if many ordinal or nominal variables are included in the data, as for every variable the researcher specifies a separate imputation model (Lüdtke et al., 2007).

Compatibility of imputation and analysis model A crucial point in the choice of a valid imputation model is compatibility with the focal analysis model that is used later on. That is, the imputation model must reflect the key structural features of this analysis model (Grund et al., 2018). If it fails to do so, parameter estimates may become biased because the imputed values no longer preserve the structure of the data. Consider, for example, the random intercept model with group means (Equation 2.4). In this setting, the imputation model must account for the nested data structure (i.e., random effects at the group level) and allow for different effects at the individual and the group level (Grund et al., 2018). **Auxiliary variables** Importantly, the imputation model is allowed to be 'richer' than the analysis model. Including additional effects or variables that are not part of the analysis model as *auxiliary variables*[5] is considered beneficial for parameter estimation (Collins et al., 2001). These variables could either be a cause of missingness, thereby making the assumption of missing at random (MAR) more plausible. Furthermore, they may correlate with variables that contain missing values, which increases the precision of the imputations and reduces bias in parameter estimates (Collins et al., 2001).

Tip

In the musical self-concept study, the years of teaching experience of the ensemble teachers might also be available. Although this variable is not part of the analysis model, it may correlate with practice behavior or musical self-concept and can therefore serve as an auxiliary level-2 variable.

In this thesis only the joint imputation will be considered. The *multivariate empty model* represents a suitable joint imputation model for the random intercept analysis model (Grund et al., 2016). Revisiting the example from the previous chapter, an imputation model with one auxiliary variable can be written as

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix} \tag{2.5}$$

Here, W_j represents a level-2 auxiliary variable and therefore has no level-1 residual term. The model preserves the clustered data structure by allowing random intercepts at the group level.

The random effects and residuals are assumed to follow multivariate normal distributions

$$\begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} \sim N(0, \Psi), \qquad \begin{pmatrix} e_{Yij} \\ e_{Xij} \end{pmatrix} \sim N(0, \Sigma)$$

where the covariance matrix Ψ captures the between-group variability and covariation of the random intercepts and Σ represents the within-group covariance structure of the level-1 residuals:

$$\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}$$

[5]Auxiliary variables are referred to as variables whose only purpose it is to improve the missing data imputation (Collins et al., 2001).

2.4.2.1.2 Analyzing the completed data sets

After the imputation step, researchers are in the comfortable position to have m completed data sets that can be analyzed using standard frequentist methods. To each of the data sets, the substantive analysis model of interest (e.g., the random intercept model) is now fitted. Each analysis proceeds according to the standard estimation procedure described in Section 2.2, up to the computation of parameter estimates and their standard errors. Hypothesis testing is not conducted at this stage. The result of this step is a set of m parameter estimates and corresponding standard errors.

2.4.2.1.3 Pooling the results

These m estimates and standard errors are aggregated into a single set of results following Rubin's pooling rules (Rubin, 1987).
He defined the point estimates of effects as the arithmetic average of the m estimates:

$$\hat{\theta} = \frac{1}{m} \sum_{k=1}^{m} \hat{\theta}_k$$

$\hat{\theta}$ is the pooled point estimate, and $\hat{\theta}_k$ is the estimate from data set k.

For the pooled standard errors, both the within-imputation variance (that is based on a complete data set) and the between-imputation variance (that adds the uncertainty due to missing values) are combined. The within-imputation variance is the arithmetic average of the imputation's variances (standard errors squared):

$$\bar{V}_w = \frac{1}{m} \sum_{k=1}^{m} SE_k^2$$

The between-imputation variance is defined as the variance of point estimates between imputations that is due to uncertainty caused by the missing values:

$$V_b = \frac{1}{m-1} \sum_{k=1}^{m} (\hat{\theta}_k - \hat{\theta})^2$$

The total pooled standard error combines the within-imputation variance $\bar{V}_w$ and the between-imputation variance V_b:

$$SE = \sqrt{\bar{V}_w + V_b + \frac{V_b}{m}}$$

The third term in this expression acts as a correction factor for using a finite number of imputations in the Monte Carlo simulation (Enders, 2022, p. 284). As the number of imputations m increases, this correction term approaches zero. As the between-imputation variance is a measure of increase in uncertainty due to missing values, the influence of missing data on the variance of estimates can be expressed proportionally, e.g. as the relative increase of variance (RIV) due to missing values:

$$RIV = \frac{V_b + \frac{V_b}{m}}{\bar{V}_w}$$

With an appropriate point estimate and standard error, the remaining step (see Figure 2.2) is the choice of the sampling distribution. Rubin proposed a calculation method for degrees of freedom of the t-distribution, that is based on the RIV and the number of imputations m:

$$df_R = (m-1)\left(1 + \frac{1}{RIV^2}\right)$$

This formula assumes that the degrees of freedom of the complete data are effectively infinite (Barnard & Rubin, 1999). In smaller samples, the classic Rubin formula therefore tends to produce degrees of freedom that are too large, sometimes even larger than the degrees of freedom of the complete data. Barnard and Rubin [@-barnard] thus proposed a small-sample-adjusted degrees-of-freedom calculation for pooled parameters. Their formula downweights the complete-data degrees of freedom to account for the missing values. As the complete-data degrees of freedom cannot easily be obtained for multilevel analyses, a suitable degrees-of-freedom approximation must be chosen for the analysis. Possible approaches include the methods described in Section 2.3.0.3, such as the Kenward–Roger correction or the simple $m - l - 1$ rule.

2.4.2.2 Full Information Maximum Likelihood (FIML)

Full information maximum likelihood (FIML) is a model-based approach in which parameter estimates are obtained directly by maximizing the likelihood of the observed data (Enders, 2022, p. 98). Under the assumption that data are missing at random (MAR), FIML yields unbiased and efficient parameter estimates (D. McNeish, 2017).

Incorporating auxiliary variables is less straightforward in FIML than in multiple imputation. Nevertheless, several approaches have been proposed to include auxiliary information in FIML estimation, such as the saturated correlates model or the extra dependent variable approach (Graham, 2003), as well as the factored regression framework described below, which forms the basis of Bayesian missing-data approaches.

For the random intercept model with group-mean effects considered in this thesis, FIML is not always well suited when missingness occurs in predictor variables. In such cases, maximum likelihood estimation requires additional distributional assumptions about the distribution of the predictors in order to incorporate them into the likelihood function (Grund et al., 2018). In multilevel models with manifest group means, these assumptions may lead to biased estimates of between-group effects (Grund et al., 2019).

2.4.2.3 Fully Bayesian Estimation

The Bayesian framework introduced in Section 2.3.1 can naturally be applied to missing-data problems. In contrast to the two-step procedure of multiple imputation, Bayesian missing-data analysis directly uses the posterior distribution of the model parameters conditional on the observed data for statistical inference.

As described in Section 2.4.2.1.1, missing values are treated as additional unknown quantities. To simulate these values during estimation, a model for the variables with missing data must be specified. A convenient way to specify the joint distribution of the outcome and the variables with missing values is to apply the probability chain rule and express the joint distribution as a product of conditional distributions, which is referred to as *sequential specification* or *factored regression specification* (Lüdtke et al., 2020):

$$g\,(y,\ x;\gamma) = g_y\left(y|x;\gamma_y\right) \times g_x(x;\gamma_x)$$

where $\gamma = (\gamma_y, \gamma_x)$ denotes the vector of all model parameters. Here, the joint density is decomposed into two components. The first term $g_y(y \mid x; \gamma_y)$ corresponds to the analysis model, describing the conditional distribution of the outcome given the covariate X. The second term $g_x(x; \gamma_x)$ specifies a model for the distribution of the covariate X. An advantage of this formulation is its flexibility: it allows the inclusion of variables with different measurement scales and facilitates the use of auxiliary variables that are not part of the substantive analysis model (Daniels et al., 2014; Erler et al., 2016). For example, auxiliary variables such as W may be included to improve the imputation of missing values while the substantive model of interest is still estimated without W. For this purpose, auxiliary variables can be placed in the first position of the factored regression, so that the analysis variables predict the auxiliary variables, but not vice versa (Enders, 2022, p. 215):

$$g\,(y,\ x, w;\gamma) = g_w\left(w|y, x;\gamma_w\right) \times g_y\left(y|x;\gamma_y\right) \times g_x(x;\gamma_x) \tag{2.6}$$

Because all missing values are resampled in each iteration, the uncertainty due to missing data is automatically included in the posterior distribution (Erler et al., 2016).

2.4.3 Present study

Of course, both challenges considered here—small sample sizes and missing data—frequently co-occur in empirical research. Some literature investigates this intersection in single-level regression. Von Hippel [@-hippel] demonstrates that multiple imputation can exhibit bias and inefficiency in small samples and that this hinges on the prior choice. McNeish [@-mcneish17] obtained good results with using joint multiple imputation with small-sample-degrees of freedom.

In a multilevel context, McNeish [@-mcneishgrowth] showed in a simulation study that MI with small-sample adjustments (REML estimation and Kenward-Rogers correction) yields better statistical properties than without corrections in a growth model. However, these adjustments were not compared to standard inferential approaches based on unadjusted degrees of freedom. More generally, this remains one of the few studies explicitly addressing the joint occurrence of small samples and missing data in multilevel settings. In particular, it remains unclear how missing values in predictor variables should be handled. In such cases, maximum likelihood estimation cannot be readily applied, whereas fully Bayesian estimation may offer a promising alternative, as it naturally accommodates both missing predictor values and small sample sizes.

To address these questions, the present study uses a simulation design to compare methods for handling missing data in the predictor variable in a small-sample multilevel random intercept model. In particular, it examines whether applying small-sample-adjusted degrees-of-freedom approximations after multiple imputation improves inferential performance, and whether fully Bayesian estimation provides a suitable alternative to imputation-based or likelihood-based approaches.

Simulation studies can be understood as computer experiments (Morris et al., 2019), in which data with known characteristics are generated and then analysed using the methods of interest. Because the true parameter values are known, the resulting estimates can be compared with the truth, allowing the performance of the methods to be evaluated (Siepe et al., 2024).

Chapter 3

Methods

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

3.1 Data generation mechanism

3.1.1 Data-generating model

The data-generating model is a two-level random intercept model:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + \gamma_{02}W_j + u_{0j} + e_{ij} \tag{3.1}$$

Y_{ij}, X_{ij} and W_j were parameterized to have a mean of zero and a variance of 1 on the population level.
First, between- and within-components of X_{ij} were simulated, independent of each other and with variances chosen to achieve a prespecified intraclass coefficient (see Section 3.1.3):

$$X_{ij} = u_{Xj} + e_{Xij},$$
$$u_{X_j} \sim N(0, {\tau_x}^2), \qquad e_{X_{ij}} \sim N(0, {\sigma_x}^2)$$

with ${\tau_x}^2 = ICC_x$ and ${\sigma_x}^2 = 1 - ICC_x$, respectively.
X_{ij} was then group-mean-centered.
The group-level covariate W_j was generated to have a prespecified correlation with the group mean of X ($\rho_{\bar{X}W}$) using a linear model[1]:

$$W_j = a\bar{X}_j + e_{Wj} \qquad e_{Wj} \sim N(0, {\sigma_w}^2)$$

Finally, the random effects on the group and the individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \qquad e_{ij} \sim N(0, \sigma^2)$$

and independent of each other.
Individual and group residual variances of Y_{ij} were set to achieve a prespecified residual intraclass correlation and $Var(Y_{ij}) = 1$, see Section A.3.

[1]Under the constraint $Var(W_j) = 1$, it follows that ${\sigma_w}^2 = 1 - \rho^2_{\bar{X}W}$ and $a = \frac{\rho_{\bar{X}W}}{\sqrt{\tau_x^2 + \frac{{\sigma_x}^2}{n}}}$, see Appendix A for details.

TABLE 3.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

Data conditions	
N_1	10
N_2	15 / 30 / 60
ICC of X and Y	.1 / .3
Correlation $\bar{X}W$	.3
Model parameters	
Effect of $(X_{ij} - \bar{X}_{.j})$	.2
Effect of $\bar{X}_{.j}$	0 / .4
Effect of W_j	0
Missing values	
Pattern	$X \sim W$
Mechanism	MCAR / MAR
Proportion	.3

3.1.2 Missing data generation

After simulation of the complete dataset, values of the X-variable have been deleted to simulate missingness. A 'missingness tendency' r was simulated for each X-value according to

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \qquad e_{ij} \sim N(0, 1 - \beta^2)$$

r_{ij} is normally distributed. Observations with $r_{ij} > 0$ were set to missing.
The intercept α controls the proportion of missingness in X, whereas the slope β determines the strength of the missing-at-random mechanism. When $\beta = 0$, missingness is completely at random (MCAR). With increasing β, the missingness mechanism becomes more dependent on the W-values; with $\beta = 1$, missingness would be completely deterministic.
For the present simulation, $\beta = 0.3$ was used for MAR conditions, corresponding to approximately 9% explained variance.

3.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table 3.1.

The number of Level-2 units was manipulated across conditions ($N_2 = 15, 30, 60$), while the number of Level-1 observations was kept constant ($N_1 = 10$). The between-cluster effect of $\bar{X}_{.j}$ was varied ($\gamma_{01} = 0, .40$), whereas the within-cluster effect of $(X_{ij} - \bar{X}_{.j})$ was held constant at .20 and the effect of W_j was set to zero. This results in conditions with either a negative and a positive context effect (i.e., $\gamma_{10} - \gamma_{01}$) of X in the population model. The ICC of X and the residual ICC of Y were varied ($\rho_X = \rho_{Y|X} = .10, .30$). Missing data were generated on X with a varying missingness mechanism (MCAR or MAR with a strength of $\beta = .30$), with a constant missingness rate of 30%. The factors were combined fully factorial, creating $3 \times 2 \times 2 \times 2 = 24$ conditions.
All in all, these settings reflect a number of the typical applications in cross-sectional and longitudinal applied research (e.g., smaller and larger intraclass correlation coefficients (ICC) and level-two sample sizes). They also create sufficient methodological variation to evaluate the properties of the methods of interest (e.g., null versus substantial effects).

3.2 Estimands of the focal analysis model

The focal analysis model is

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + u_{0j} + e_{ij} \tag{3.2}$$

The primary target is the Level-2-effect, γ_{01}. Additionally, the Level-1-effect γ_{10} was evaluated.

3.3 Methods of Missing Data Handling

Four methods will be compared, alongside the benchmark analysis of Complete Data (CD).

3.3.1 Listwise Deletion (LD)

Listwise Deletion was used as a standard reference method for missing data handling. Cases with missing X-values were excluded prior to analysis. The group means $\bar{X}_{.j}$ were then calculated only from the remaining observations. Then, group-mean centering was applied.

3.3.2 Multiple Imputation (MI) with standard (MI-R) and adjusted (MI-a) degrees of freedom

For both multiple imputation methods, a joint modeling approach was employed. The imputation model was the multivariate empty model specified before (Equation 2.5), with W_j as an auxiliary variable on level 2.

Imputation was executed with the R package `jomo` (Quartagno & Carpenter, 2023) with 2000 burn-in-iterations and 3000 post-burn-in iterations, saving every 300th imputation (10 imputations).
For MI-R, degrees of freedom for calculating the test statistic were calculated with the software standard (Rubin´s rules, assuming infinite degrees of freedom in complete data).
For MI-a, complete data degrees of freedom were calculated with $df_{\gamma_{10}} = M - r - 1$ (M: total number of Level-1 units; r: total number of predictors, in the present case 2) and $df_{\gamma_{01}} = N_2 - q - 1$ (N_2: number of Level-2 units; q: number of Level-2 predictors, here: 1).

3.3.3 Fully Bayesian Analysis

A factored regression specification with the auxiliary variable W was used within the Bayesian framework that corresponds to the model described in Equation 2.6.

The burn-in period was set to 1000 iterations, followed by 3000 post-burn-in iterations. The model was estimated using the package `mdmb` (Robitzsch & Luedtke, 2024) with its default noninformative priors.

3.3.4 Extracted quantities

For complete data, listwise deletion and both multiple imputation methods, point estimates of γ_{10} and γ_{01} as well as their standard errors and 95% - confidence intervals were extracted.

For the fully Bayesian method, the posterior means of γ_{10} and γ_{01} were used as point estimates along with posterior standard deviations and 95% credible intervals. For comparability with the frequentist approaches and in line with common simulation practice, these posterior summaries were treated analogously to frequentist estimates and intervals (Morris et al., 2019).

3.4 Performance measures

Three performance measures were used for each parameter (γ_{10} and γ_{01}):

1. Bias of point estimates, defined as the difference between the parameter´s mean estimated and true value.
2. Coverage of confidence/credible intervals, defined as the proportion of simulation repetitions that captured the true parameter value inside the CI.
3. Standard deviation of the estimations across simulation repetitions.

Monte Carlo uncertainty of the estimated performance measures was calculated using Monte Carlo Standard Errors (MCSE), using the formulas provided in Siepe et al. (2024).
For each condition, 1000 simulation replications were used, based on comparable simulation studies (Enders et al., 2016; Grund et al., 2018) and ensuring a sufficiently small MCSE of $< 1\%$ for coverage [2].

The simulation was conducted in RStudio with the aim of full reproducibility, following guidelines of the repro book (ZITIEREN), e.g. by setting a seed for simulation and using reproducible package environments. The code and all relevant information for reproduction can be found in the digital version of the master thesis. (*LINK*)

Noch einfügen: wie Performance measures evaluiert werden sollen; außerdem width of confidence intervals als Performance measure?

[2] MCSE calculation for bias and standard deviation are based on the sample variance of the estimates and cannot be pre-calculated.

Chapter 4

Results

Convergence diagnostics for the MCMC algorithm were evaluated under an exemplary critical condition and indicated satisfactory convergence and autocorrelation behavior (see Appendix B).
As expected, no substantial or noteworthy patterns were observed for the individual-level effect γ_{10}; therefore, detailed results are reported here only for γ_{01}. Full results for all parameters of interest are provided in Appendix C.

4.1 Bias of point estimation

The findings for γ_{01} are depicted in Figure 4.1. Listwise deletion produced substantial bias, especially in low-ICC conditions, with positive bias for the null effect and negative bias for the positive between-group effect. With increasing sample size and under MAR condition, this bias was more pronounced.
Multiple imputation showed even larger bias than listwise deletion in conditions with low ICC and positive effect[1]. In contrast, the Bayesian approach only led to noticeable bias in the most challenging conditions (smallest sample size, low ICC and positive γ_{01}-effect).

4.2 Coverage of confidence/credible intervals

The coverage of the confidence and credible intervals for γ_{01} is shown for $N_2 = 15$ in Figure 4.2, as coverage for larger sample sizes was largely within the acceptable range of 0.925–0.975.

Listwise deletion yielded coverage rates close to the nominal 95% level across all conditions. For multiple imputation, the small-sample adjusted degrees of freedom resulted in increased overcoverage compared to the standard degrees of freedom, particularly when $N_2 = 15$, with a similar but weaker tendency at larger sample sizes. The Bayesian approach showed even stronger overcoverage in the smallest sample, but not for $N_2 = 30$ and $N_2 = 60$.

Overall, MI with standard degrees of freedom (MI-R) and listwise deletion provided coverage closest to the nominal level, whereas noticeable deviations occurred mainly in the smallest Level-2 sample for the other methods.

[1]However, in a sensitivity analysis, usage of alternative weakly informative priors in these conditions consistently reduced bias by roughly 30–60% (see Appendix D for details).

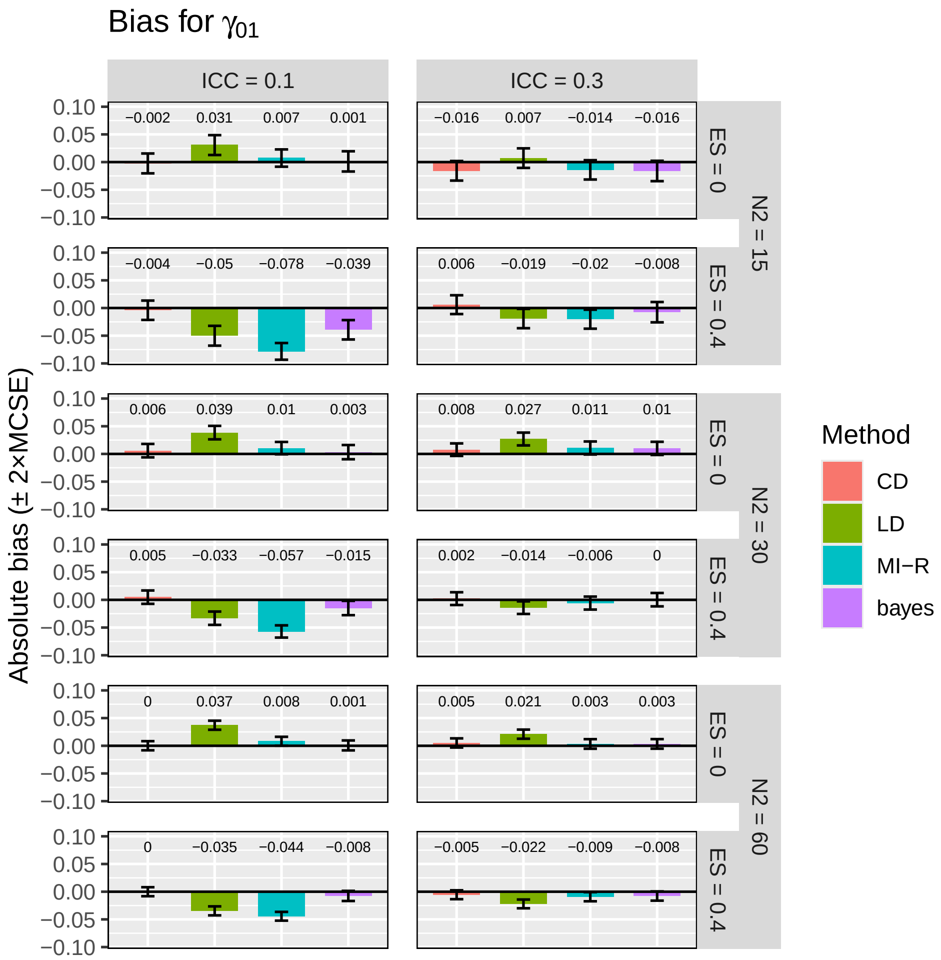


FIGURE 4.1: Hier noch Caption

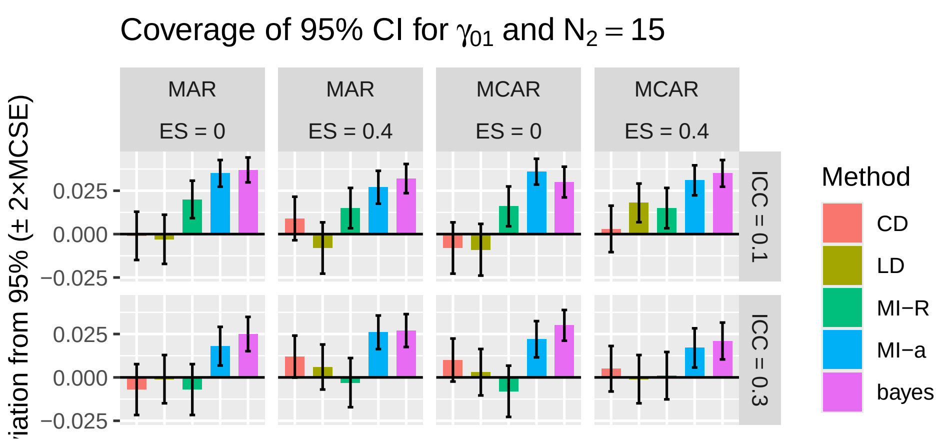


FIGURE 4.2: Coverage TEXT TEXT TEXT HIER HIN NOCH TEXT

4.3 Standard deviation of point estimates

No substantial differences between methods for the parameter γ_{10} were observed, except for the complete-data condition, which consistently yielded the smallest standard deviations.
Interestingly, for γ_{10}, the variability of point estimates across simulation repetitions was smallest for multiple imputation in low-ICC conditions and similar to complete data analysis for high-ICC, whereas the Bayesian approach produced the largest standard deviations across all conditions (see Figure 4.3). A systematic tendency was visible: Multiple imputation standard deviations decreased with a low ICC, whereas in the Bayesian approach, they decreased with a high ICC for $N_2 = 30$ and $N_2 = 60$. Listwise deletion shows very similar results to complete data analysis regarding standard deviation.
Differences between methods are overall quite small, with MI SD being smaller than Bayes SD by a maximum of ~14.5%.

Hier eher Tabelle statt Grafik?

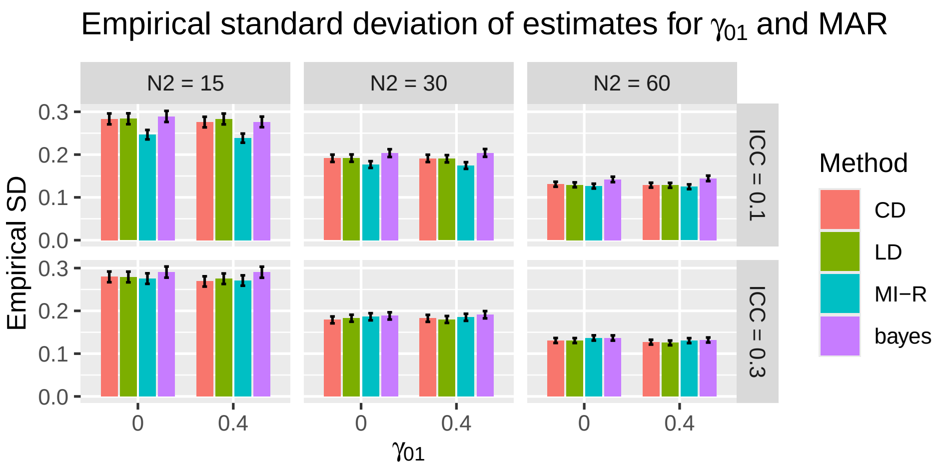


FIGURE 4.3: blablabalbslsd

4.4 Width of confidence / credible intervals

For γ_{01}, interval widths were relatively stable within each sample size, but there were differences between methods. Bayesian credible intervals were consistently the widest, followed by the MI-a approach. Confidence intervals from the MI with Rubin´s degrees of freedom were even narrower than CD/LD for $N_2 = 15$, although these differences were small in magnitude.
These results are shown in an exemplary manner in Figure 4.4.

4.5 Summary of results

Regarding γ_{01}, listwise deletion showed increased bias but acceptable coverage. Multiple imputation led to higher bias than the Bayesian approach in several challenging conditions. MI-a produced higher overcoverage and wider confidence intervals compared to the MI-R variant; the Bayesian method generally resulted in the highest overcoverage and the widest credible intervals.

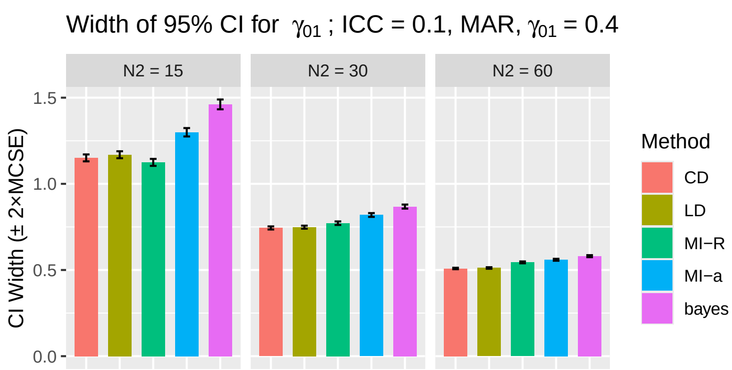


Figure 4.4: blablabalbslsd

Chapter 5

Discussion

5.1 The role of small-sample-adjusted degrees of freedom in MI

A misspecification of degrees of freedom — such as assuming infinite complete-data degrees of freedom instead of approximately 13 in the smallest condition at level 2 — would intuitively be expected to result in undercoverage and overly narrow confidence intervals. However, in the present simulation, coverage remained nominal or was even slightly above the nominal level despite this misspecification. This suggests that other sources of uncertainty within the multiple imputation framework may play a more dominant role in determining interval width and coverage. Overcoverage was furthermore not only observed in the multiple imputation framework, but also in the fully Bayesian approach, where it was even more pronounced. This emphasizes that the conservative behavior is not specific to Rubin's rules or the pooling procedure, but rather reflects properties of the underlying MCMC algorithm.
Under the conditions investigated, a small-sample adjustment of degrees of freedom in multiple imputation for random intercept models with manifest group mean predictors may therefore not be critically important. This finding may provide some reassurance for researchers relying on default software implementations.

5.2 Other explanations for the results

The average model-based standard errors consistently exceeded the empirical standard deviation of the estimates, indicating that variability was overestimated. As confidence interval width is directly determined by these standard errors, this provides a straightforward explanation for the observed overcoverage. Joint modeling with `jomo` was also found to be conservative in another study (Audigier et al., 2018). The authors state there that the use of inverse Wishart priors with few degrees of freedom introduces too much variability in fixed effect estimates.

Andere Erklärungen:

5.2.1 Schwacher MAR-Mechanismus

eher wenig Struktur die Missingness erklärt (9% Varianz)-> Imputationsmodell "rät" mehr –> hohe Varianz in der Imputation –> hohe Varianz beim Pooling between-imputations

–> aber diese Unsicherheit gibt es ja wirklich, sie ist also nicht "zu" hoch, oder?

5.2.2 Missingness in X

Da die Parameter ja mithilfe von X geschätzt werden, sind sie unsicher, wenn X unsicher ist (und das wird ja durch fehlende Werte höher) -> ist das ein Grund?

kleiner ICC-> wenig Varianz auf Level 2

5.2.3 Prior choice

die Priors lassen zu viel Varianz zu, sodass die Imputationen sehr weit streuen; weiter als die "echten" Werte streuen würden (?)

The more pronounced overcoverage observed in the fully Bayesian analyses may also be related to the prior specification used in `mdmb`. Their default priors allow for even more variability in the parameter estimates than in `jomo`.

These findings suggest that the degree of prior regularization plays an important role in determining inferential properties such as coverage.

5.3 A trade-off between bias and variance

Wie entsteht der Bias in MI eigentlich? Vielleicht dadurch, dass die group mean aggregation erst nach der Imputation durchgeführt wird? Ist das dadurch "unsauber", verzerrt? ...

MI Diskussion Priors ICC .4 gewinnt doppelt:

Prior passt besser

mehr echte Information (weil weniger within-Varianz)

5.4 Limitations and further research

nur geringes MAR, nur kontinuierliche Variablen, keine Random Slopes, einfaches Modell, keine geringsten Gruppengrößen, immer gleich große Gruppen (macht das was aus?), kein MNAR, keine priors angeguckt oder manipuliert, simulierte Daten die die Normalitätsannahmen erfüllen

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Appendix A

Derivations of equations of data-generating model

A.1 Calculation of σ_w^2

The data-generating model for W_j is:

$$W_j = a \cdot \bar{X}_j + e_{Wj}, \qquad e_{Wj} \sim N(0, {\sigma_w}^2) \tag{A.1}$$

The variance of W_j is therefore:

$$Var(W_j) = Var(a \cdot \bar{X}_j + e_{Wj}) = a^2 \cdot Var(\bar{X}_j) + Var(e_{Wj}) \tag{A.2}$$

Substituting the formula for slope calculation $a = \frac{Cov(W_j,\ \bar{X}_j)}{Var(\bar{X}_j)}$ into Equation A.2 yields:

$$\begin{aligned} Var(W_j) &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)^2} \cdot Var(\bar{X}_j) + Var(e_{Wj}) \\ &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \end{aligned}$$

This term can be simplified using $Cov(A, B) = Corr(A, B) \cdot SD(A) \cdot SD(B)$:

$$\begin{aligned} Var(W_j) &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= \frac{\left(Corr(W_j, \bar{X}_j) \cdot SD(W_j) \cdot SD(\bar{X}_j)\right)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= \frac{Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) \cdot Var(\bar{X}_j)}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) + Var(e_{Wj}) \end{aligned}$$

Thus,

$$
\begin{aligned}
Var(W_j) &= Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) + Var(e_{Wj}) \\
Var(e_{Wj}) &= Var(W_j) - Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) \\
&= Var(W_j)\left(1 - Corr(W_j, \bar{X}_j)^2\right)
\end{aligned}
$$

Since $Var(W_j) = 1$:

$$Var(e_{Wj}) = 1 - Corr(W_j, \bar{X}_j)^2$$

In short notation:

$$\sigma_w^2 = 1 - \rho_{\bar{X}W}^2$$

A.2 Calculation of the slope to generate W_j

$$a = \frac{Cov(W_j, \bar{X}_j)}{Var(\bar{X}_j)}$$

The variance of $\bar{X}_j$ can be derived as follows:

$$
\begin{aligned}
Var(\bar{X}_j) &= Var\left(\frac{1}{n}\sum_{i=1}^{n} X_{ij}\right) \\
&= Var\left(\frac{1}{n}\sum_{i=1}^{n}(u_{Xj} + e_{Xij})\right) \\
&= Var\left(\frac{1}{n}\sum_{i=1}^{n} u_{Xj} + \frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var\left(u_{Xj} + \frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + Var\left(\frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2}Var\left(\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2}\sum_{i=1}^{n} Var(e_{Xij}) \\
&= \tau_x^2 + \frac{n \cdot \sigma_X^2}{n^2} \\
&= \tau_x^2 + \frac{\sigma_X^2}{n}
\end{aligned}
$$

where n denotes the group size (cluster size), assumed to be constant across groups.\
The covariance $Cov(W_j, \bar{X}_j)$ is therefore:

$$\begin{aligned} Cov(W_j, \bar{X}_j) &= \rho_{\bar{X}W} \cdot SD(\bar{X}_j) \cdot SD(W_j) \\ &= \rho_{\bar{X}W} \cdot \sqrt{Var(\bar{X}_j)} \cdot \sqrt{Var(W_j)} \\ &= \rho_{\bar{X}W} \cdot \sqrt{\tau_x^2 + \frac{\sigma_X^2}{n}} \cdot 1 \end{aligned}$$

A.3 Variances of random components of Y

Individual and group residual variances of Y ($Var\left(u_{0j}\right) = \psi^2$ and $Var(e_{ij}) = \sigma^2$) are determined to obtain a prespecified residual ICC of Y:

$$ICC_{Y|X,W} = \frac{\psi^2}{\psi^2 + \ \sigma^2}$$

Because $Var(Y) = 1$ and residuals are uncorrelated with predictors and with each other,

$$\psi^2 + \sigma^2 = Var(\text{Residual}) = 1 - Var(\text{fixed})$$

where $Var(\text{fixed})$ denotes the variance explained by the fixed effects of the model.\ The variance components can therefore be expressed as:

$$Var(u_{0j}) = ICC_{Y|X,W} \cdot (1 - Var(\text{fixed}))$$

$$Var(e_{ij}) = (1 - ICC_{Y|X,W}) \cdot (1 - Var(\text{fixed}))$$

$Var(\text{fixed})$ can be calculated as:

$$Var(\text{fixed}) = Var(\gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + \gamma_{02}W_j)$$

The terms $(X_{ij} - \bar{X}_{.j})$ and $\bar{X}_{.j}$ are uncorrelated. W_j is correlated with $\bar{X}_{.j}$ but not with X_{ij} (see Equation Equation A.1), and $Var(W_j)$ is set to 1, which leads to:

$$Var(\text{fixed}) = \gamma_{10}^2 \cdot Var(X_{ij} - \bar{X}_{.j}) + \gamma_{01}^2 \cdot Var(\bar{X}_{.j}) + \gamma_{02}^2 + 2\gamma_{01}\gamma_{02}Cov(\bar{X}_{.j}, W_j)$$

Using the decomposition of X_{ij} introduced in Section Section 3.1.1,

$$X_{ij} = u_{Xj} + e_{Xij},$$

the variance of $(X_{ij} - \bar{X}_{.j})$ can be derived as follows:

$$
\begin{aligned}
Var(X_{ij} - \bar{X}_{.j}) &= Var(X_{ij}) + Var(\bar{X}_{.j}) - 2 \cdot Cov(X_{ij}, \bar{X}_{.j}) \\
&= Var(X_{ij}) + Var(\bar{X}_{.j}) - 2 \cdot \Bigg(Var(u_{Xj}) + \frac{1}{n} \sum_{i'=1}^{n} Cov(u_{Xj}, e_{Xi'j}) \\
&\qquad + Cov(e_{Xij}, u_{Xj}) + \frac{1}{n} \sum_{i'=1}^{n} Cov(e_{Xij}, e_{Xi'j}) \Bigg) \\
&= Var(X_{ij}) + Var(\bar{X}_{.j}) - 2 \cdot \left(Var(u_{Xj}) + \frac{1}{n} Var(e_{Xij}) \right) \\
&= Var(X_{ij}) + Var(\bar{X}_{.j}) - 2 \cdot Var(\bar{X}_{.j}) \\
&= Var(X_{ij}) - Var(\bar{X}_{.j}) \\
&= \tau_x^2 + \sigma_X^2 - \left(\tau_x^2 + \frac{\sigma_X^2}{n} \right) \\
&= \sigma_X^2 - \frac{\sigma_X^2}{n}
\end{aligned}
$$

Because γ_{02} is set to 0, $Var(\text{fixed})$ can be calculated as

$$
Var(\text{fixed}) = \gamma_{10}^2 \left(\sigma_X^2 - \frac{\sigma_X^2}{n} \right) + \gamma_{01}^2 \left(\tau_x^2 + \frac{\sigma_X^2}{n} \right).
$$

The variances of the random components can thus be calculated like this:

$$
\begin{aligned}
\psi^2 &= ICC_{Y|X,W} \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n_j}) \\
\sigma^2 &= (1 - ICC_{Y|X,W}) \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n_j})
\end{aligned}
$$

Appendix B

Convergence analysis for MI and fully Bayesian MCMC

Convergence behaviour was tested in one run of the most challenging condition ($N_2 = 15, \gamma_{01} = .4, \beta = .3, ICC = .1$).

B.1 Diagnostics of multiple imputation

Convergence of the imputation model was assessed using potential scale reduction factors (Gelman & Rubin, 1992). Scale reduction factors that are substantially larger than 1 indicate that convergence was not yet reached fully; an acceptable threshold can be set at 1.1 (Gelman et al., 2025, pp. 285–287), while a stricter recommendation was recently set to be 1.01 (Vehtari et al., 2021). Given the simulation setting with aggregated results over many replications, the first threshold was considered sufficient.

All values were close to 1 (maximum $\hat{R} = 1.026$), indicating satisfactory convergence across all parameters.

Trace plots showed no signs of substantial change after burn-in. Inspection of the autocorrelation plots indicated that autocorrelation diminished quickly across increasing lags and did not persist over longer distances between samples. The distance of 300 lags between imputation that was chosen in this simulation was therefore sufficient. Diagnostic plots are shown in Figure B.1 for the most critical parameter, the level-2 residual variance of Y.

B.2 Diagnostics of fully Bayesian analysis

$\hat{R}$-values were close to 1, except for the variance of X on level 2 ($\hat{R} = 1.05$) and the regression coefficient of W on the group mean of X ($\hat{R} = 1.03$).

Trace plots showed no drifting, and autocorrelation plots indicated sufficiently low autocorrelation after few lags. They are shown here for the two parameters with the highest potential scale reduction factor, as well as for τ_0^2 as it is a critical parameter in this simulation.

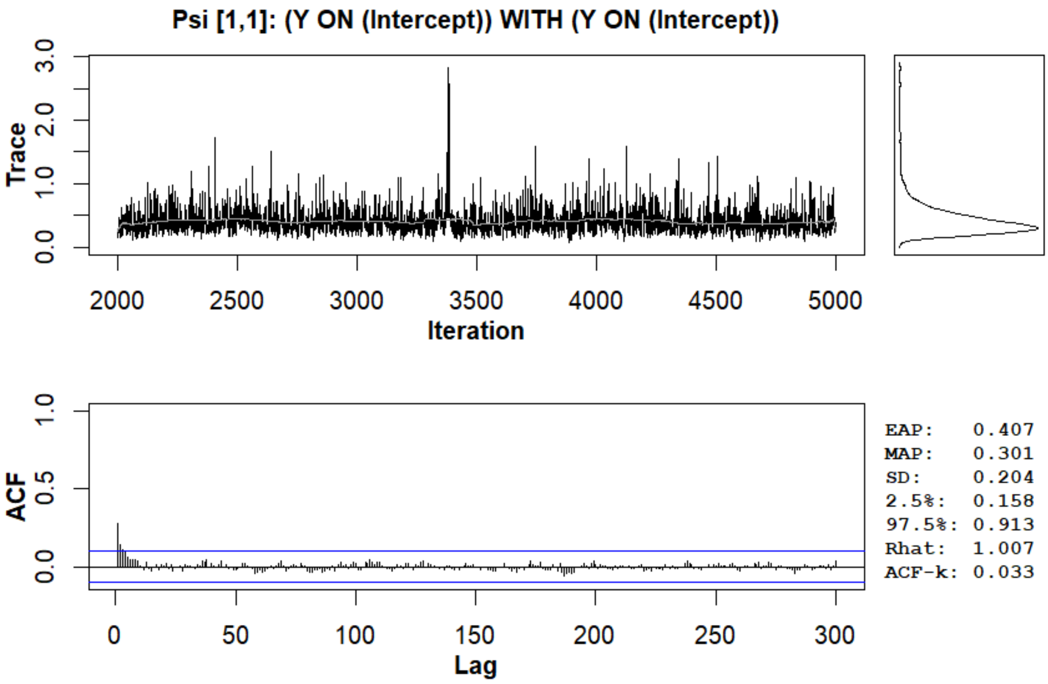

FIGURE B.1: Trace and autocorrelation plot in an exemplary simulation run for the level-2 covariance τ_0^2 in MI

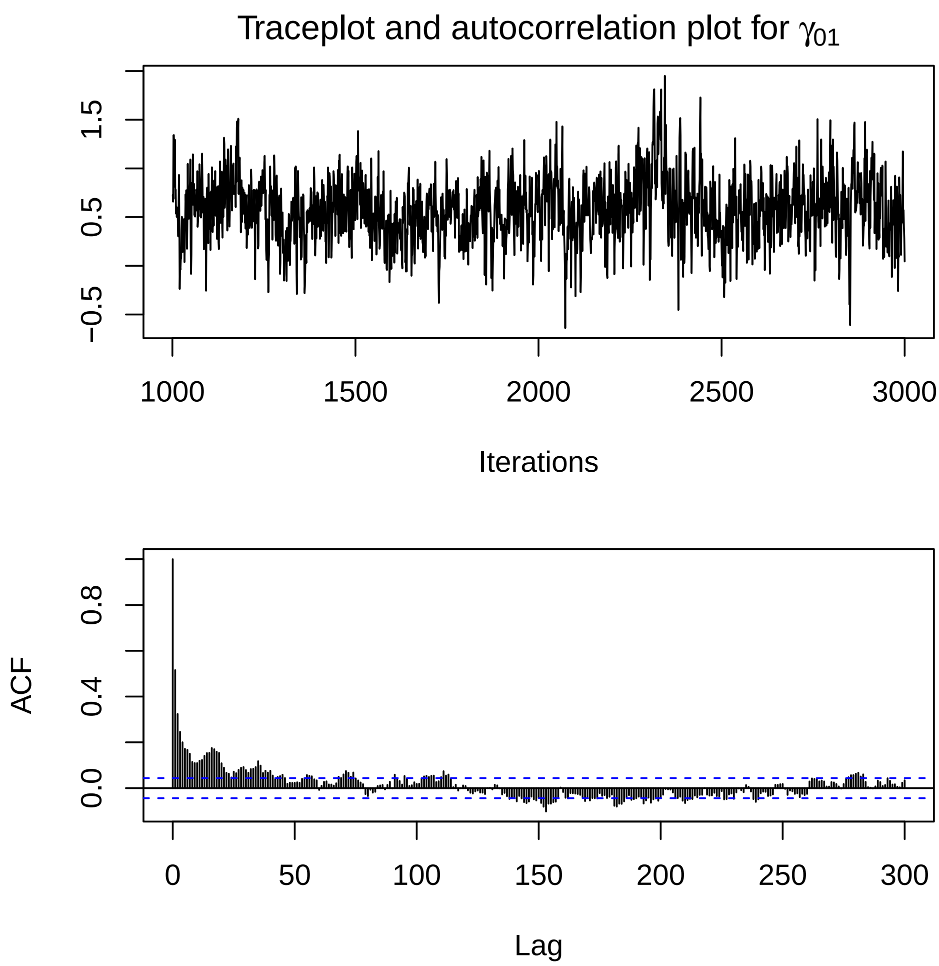

FIGURE B.2: Trace and autocorrelation plot in an exemplary simulation run for the level-2 effect γ_{01} in fully Bayesian analysis

Appendix C

Full summary of results for the performance measures

TABLE C.1: Simulation results (N2 = 15, $\gamma_{01} = 0.0$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.002	0.942	0.284	0.037	0.941	0.300	0.005	0.966	0.252	0.005	0.986	0.252	0.003	0.980	0.291
γ_{10}	-0.006	0.951	0.084	-0.006	0.944	0.104	-0.008	0.943	0.103	-0.008	0.944	0.103	-0.010	0.946	0.100
ICC = 0.3															
γ_{01}	0.000	0.960	0.272	0.012	0.953	0.280	-0.001	0.942	0.278	-0.001	0.972	0.278	-0.003	0.980	0.288
γ_{10}	0.001	0.942	0.089	0.000	0.931	0.111	-0.005	0.938	0.108	-0.005	0.939	0.108	-0.006	0.934	0.108
MAR															
ICC = 0.1															
γ_{01}	-0.002	0.949	0.283	0.031	0.947	0.284	0.007	0.970	0.247	0.007	0.985	0.247	0.001	0.987	0.289
γ_{10}	0.000	0.942	0.087	-0.002	0.954	0.106	-0.007	0.949	0.104	-0.007	0.950	0.104	-0.007	0.954	0.102
ICC = 0.3															
γ_{01}	-0.016	0.943	0.279	0.007	0.949	0.279	-0.014	0.943	0.275	-0.014	0.968	0.275	-0.016	0.975	0.291
γ_{10}	0.001	0.941	0.087	-0.001	0.948	0.105	-0.004	0.943	0.102	-0.004	0.945	0.102	-0.005	0.949	0.101

Table C.2: Simulation results (N2 = 15, γ_{01} = 0.4)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	-0.008	0.953	0.275	-0.038	0.968	0.269	-0.075	0.965	0.239	-0.075	0.981	0.239	-0.036	0.985	0.281
γ_{10}	0.000	0.952	0.083	0.000	0.952	0.102	-0.001	0.950	0.101	-0.001	0.953	0.101	-0.003	0.954	0.099
ICC = 0.3															
γ_{01}	0.003	0.955	0.272	-0.013	0.949	0.272	-0.016	0.951	0.275	-0.016	0.967	0.275	-0.008	0.971	0.290
γ_{10}	0.002	0.949	0.083	0.003	0.935	0.106	-0.001	0.944	0.103	-0.001	0.948	0.103	-0.003	0.942	0.101
MAR															
ICC = 0.1															
γ_{01}	-0.004	0.959	0.276	-0.050	0.942	0.283	-0.078	0.965	0.238	-0.078	0.977	0.238	-0.039	0.982	0.276
γ_{10}	-0.007	0.954	0.084	-0.005	0.952	0.105	-0.007	0.956	0.102	-0.007	0.959	0.102	-0.010	0.952	0.101
ICC = 0.3															
γ_{01}	0.006	0.962	0.269	-0.019	0.956	0.275	-0.020	0.947	0.271	-0.020	0.976	0.271	-0.008	0.977	0.290
γ_{10}	-0.001	0.945	0.086	0.000	0.948	0.105	-0.003	0.946	0.102	-0.003	0.948	0.102	-0.006	0.951	0.102

TABLE C.3: Simulation results (N2 = 30, γ_{01} = 0.0)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.003	0.945	0.188	0.042	0.939	0.186	0.014	0.962	0.173	0.014	0.976	0.173	0.009	0.963	0.199
γ_{10}	-0.005	0.953	0.059	-0.006	0.953	0.073	-0.007	0.954	0.073	-0.007	0.954	0.073	-0.008	0.950	0.072
ICC = 0.3															
γ_{01}	-0.004	0.957	0.185	0.010	0.964	0.185	-0.003	0.955	0.190	-0.003	0.964	0.190	-0.003	0.965	0.192
γ_{10}	0.001	0.942	0.063	0.000	0.943	0.077	-0.003	0.939	0.077	-0.003	0.940	0.077	-0.003	0.943	0.076
MAR															
ICC = 0.1															
γ_{01}	0.006	0.941	0.191	0.039	0.947	0.192	0.010	0.965	0.177	0.010	0.972	0.177	0.003	0.965	0.203
γ_{10}	0.000	0.944	0.061	-0.002	0.945	0.075	-0.003	0.950	0.074	-0.003	0.951	0.074	-0.004	0.948	0.073
ICC = 0.3															
γ_{01}	0.008	0.969	0.179	0.027	0.956	0.183	0.011	0.958	0.186	0.011	0.968	0.186	0.010	0.967	0.188
γ_{10}	0.000	0.944	0.062	0.000	0.948	0.076	-0.003	0.954	0.075	-0.003	0.954	0.075	-0.003	0.946	0.074

Table C.4: Simulation results (N2 = 30, γ_{01} = 0.4)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.005	0.950	0.189	-0.034	0.956	0.187	-0.052	0.956	0.175	-0.052	0.966	0.175	-0.009	0.970	0.204
γ_{10}	0.000	0.966	0.056	-0.001	0.967	0.068	0.001	0.966	0.066	0.001	0.969	0.066	-0.003	0.967	0.065
ICC = 0.3															
γ_{01}	0.007	0.943	0.188	-0.008	0.941	0.191	-0.002	0.938	0.194	-0.002	0.953	0.194	0.004	0.951	0.201
γ_{10}	0.000	0.963	0.057	0.003	0.956	0.069	0.001	0.950	0.067	0.001	0.954	0.067	0.000	0.954	0.067
MAR															
ICC = 0.1															
γ_{01}	0.005	0.948	0.191	-0.033	0.944	0.190	-0.057	0.952	0.174	-0.057	0.963	0.174	-0.015	0.958	0.204
γ_{10}	0.001	0.947	0.060	0.001	0.946	0.073	0.003	0.948	0.073	0.003	0.950	0.073	0.000	0.946	0.071
ICC = 0.3															
γ_{01}	0.002	0.941	0.182	-0.014	0.947	0.180	-0.006	0.942	0.185	-0.006	0.951	0.185	0.000	0.954	0.191
γ_{10}	-0.002	0.948	0.059	-0.001	0.947	0.075	-0.003	0.950	0.072	-0.003	0.951	0.072	-0.004	0.948	0.072

Table C.5: Simulation results (N2 = 60, $\gamma_{01} = 0.0$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	-0.006	0.955	0.130	0.027	0.954	0.128	-0.002	0.971	0.124	-0.002	0.979	0.124	-0.010	0.961	0.139
γ_{10}	0.000	0.944	0.043	-0.001	0.953	0.053	-0.002	0.939	0.053	-0.002	0.940	0.053	-0.002	0.945	0.052
ICC = 0.3															
γ_{01}	-0.002	0.946	0.127	0.017	0.955	0.127	0.001	0.949	0.132	0.001	0.954	0.132	0.000	0.954	0.133
γ_{10}	0.000	0.947	0.042	0.002	0.947	0.052	0.000	0.948	0.051	0.000	0.948	0.051	0.000	0.946	0.051
MAR															
ICC = 0.1															
γ_{01}	0.000	0.951	0.131	0.037	0.949	0.129	0.008	0.960	0.126	0.008	0.965	0.126	0.001	0.953	0.142
γ_{10}	-0.001	0.954	0.043	0.000	0.954	0.052	-0.001	0.956	0.051	-0.001	0.957	0.051	-0.001	0.953	0.050
ICC = 0.3															
γ_{01}	0.005	0.948	0.131	0.021	0.949	0.131	0.003	0.946	0.137	0.003	0.949	0.137	0.003	0.956	0.137
γ_{10}	0.001	0.954	0.041	0.000	0.949	0.052	-0.001	0.947	0.052	-0.001	0.947	0.052	-0.001	0.944	0.051

Table C.6: Simulation results (N2 = 60, γ_{01} = 0.4)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.001	0.946	0.126	-0.039	0.939	0.124	-0.044	0.955	0.121	-0.044	0.960	0.121	-0.008	0.952	0.136
γ_{10}	0.001	0.960	0.041	0.001	0.961	0.050	0.004	0.954	0.049	0.004	0.955	0.049	0.001	0.958	0.049
ICC = 0.3															
γ_{01}	-0.002	0.952	0.129	-0.021	0.947	0.128	-0.007	0.954	0.132	-0.007	0.957	0.132	-0.005	0.962	0.133
γ_{10}	-0.001	0.954	0.040	-0.001	0.956	0.049	-0.002	0.956	0.048	-0.002	0.956	0.048	-0.002	0.961	0.048
MAR															
ICC = 0.1															
γ_{01}	0.000	0.954	0.129	-0.035	0.943	0.128	-0.044	0.960	0.125	-0.044	0.962	0.125	-0.008	0.957	0.144
γ_{10}	0.002	0.926	0.043	0.001	0.942	0.051	0.003	0.941	0.050	0.003	0.942	0.050	0.001	0.941	0.050
ICC = 0.3															
γ_{01}	-0.005	0.950	0.127	-0.022	0.956	0.125	-0.009	0.958	0.130	-0.009	0.961	0.130	-0.008	0.962	0.132
γ_{10}	-0.001	0.953	0.042	-0.002	0.944	0.051	-0.003	0.944	0.050	-0.003	0.945	0.050	-0.003	0.942	0.050

Appendix D

Default priors and sensitivity analysis

D.1 Default priors in `mdmb`

Default priors in `mdmb` are noninformative. For regression coefficients, a uniform prior on the interval $(-\infty, \infty)$ is approximated by a normal distribution with large variance, i.e., $\mathcal{N}(0, 10^{10})$. For variance–covariance matrices, improper inverse-Wishart priors are used, i.e., $\mathcal{IW}(\mathbf{0}, -p-1)$ (Asparouhov & Muthen, 2010).

D.2 Default priors in `jomo`

In `jomo`, inverse-Wishart priors are used for the covariance matrices on level 1 and 2 for imputation:

$$\Sigma \sim \mathcal{IW}(\nu, \mathbf{S})$$

where
- ν denotes the degrees of freedom, and
- $\mathbf{S}$ is the scale matrix.

The default is set to 'least informative' priors with largest dispersion, i.e., smallest possible degrees of freedom, scaled by the identity matrix (Grund et al., 2015; Quartagno & Carpenter, 2023). For the three variables of imputation model (X, Y, Z):

$$\mathbf{S} = \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

D.3 Sensitivity analysis with weakly informative priors in `jomo`

In this analysis, for the conditions when multiple imputation produced the highest bias, i.e. when $ICC = 0.1$ and $\gamma_{01} = .4$, alternative inverse Wishart priors were specified for the level-2 covariance matrix. This was done by specifying an informed guess of the covariance matrix. In the simulation study, this guess was based on the true parameter values; in empirical applications, it could be derived from previous

Table D.1: Bias reduction with alternative priors

N2	Bias.d	rel. Bias.d	Bias.a	rel. Bias.a	Bias red.
MCAR					
15	-0.075	18.7%	-0.048	12.1%	35.5%
30	-0.062	15.4%	-0.033	8.3%	46.2%
60	-0.040	9.9%	-0.014	3.6%	63.7%
MAR					
15	-0.068	17.1%	-0.037	9.1%	46.6%
30	-0.059	14.8%	-0.033	8.2%	44.3%
60	-0.045	11.2%	-0.021	5.3%	52.3%

Note:
d = default priors, a = alternative priors, Bias red. = Bias reduction

Table D.2: Uncertainty measures with alternative priors

N2	Covd	Cova	SEd	SEa	empSEd	empSEa
MCAR						
15	0.967	0.963	0.288	0.301	0.240	0.257
30	0.959	0.964	0.197	0.203	0.172	0.184
60	0.951	0.954	0.138	0.142	0.129	0.136
MAR						
15	0.954	0.965	0.286	0.300	0.251	0.266
30	0.955	0.965	0.198	0.205	0.180	0.191
60	0.965	0.966	0.139	0.142	0.120	0.128

Note:
Note. Cov = coverage; SE = model-based standard error; empSE = empirical standard error.

research. This guess for the covariance matrix corresponds to the two standardized level-1 variables X and Y (ICC = .1) and the standardized level-2 variable W.

$$\tau_{\text{guess}} = \begin{bmatrix} .1 & 0 & 0 \\ 0 & .1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

τ denotes the level-2 covariance matrix of the random effects. The level-2 covariance prior was then defined as

$$\tau \sim \mathcal{IW}(\nu, \mathbf{S}), \qquad \mathbf{S} = 3\,\tau_{\text{guess}}$$

Only the scale matrix was modified, while the degrees of freedom were kept at their default value, such that the prior remained weakly informative but was centered around more plausible variance values. The level-1 covariance matrix was kept at the default specification.

Summarized results of this analysis can be found in Table D.1.